



SUPPLY CHAIN CHALLENGES FOR SMES ENTERING EMERGING MARKETS IN SOUTH ASIA

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Abstract / Executive Summary

This research paper investigates the supply chain issues that are encountered by SMEs operating in emerging economies in South Asia. The focus is on the effects of inadequate infrastructure, financial, regulatory, and technology-related issues on SMEs and on the role of digitalisation in increasing the performance of their supply chains.

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1. Introduction

1.1. Background

Small and Medium-sized Enterprises (SMEs) that are entering the emerging markets in South Asian countries, such as *India*, *Sri Lanka*, the *Maldives*, *Pakistan*, and *Bangladesh*, are facing various issues regarding supply chain management (SCM). Issues such as logistic volatility, due to breakdowns in the supply chain, given the bad transportation system and roads, affect the process heavily. Apart from that, logistics costs in emerging markets such as India and Sri Lanka are restricting the development of SCM in South Asia (Ghosh *et al.*, 2026). Apart from this, the small and medium-sized enterprises supply chain (SMESC) system requires a huge amount of resources, especially financial resources and time for warehouse upgrades, technology adoption, or inventory management systems. This leads to higher interest rates, taxation, and inadequacy in working capital. Therefore, the organisations are often forced to take loans from the banks, which prevent them from achieving economies of scale in the business process, leading to slower growth and expansion (Vinh *et al.*, 2025).

Since the introduction of Industry 4.0, digital technology has become an essential part of any business operation, incorporating aspects such as Internet of Things (IoT), big data analytics, and cloud computing. However, given the challenges regarding this are data integration, cybersecurity, high upfront investment, weak policy, and business risks (Alshahrani, 2023). Apart from that, in the context of implementing blockchain technology, the support of the government is essential for the efficacy of the system. A lack of skilled human resources and support from the government in the form of financial subsidy and legal aid through specific acts helps an SME to achieve growth and efficiency. Moreover, SMEs heavily rely on local suppliers, niche consumer bases, and often boutique-level service providers, and these make them highly vulnerable to bankruptcy, raw material shortage, and quality issues (Bui and Le, 2025).

Another major issue regarding the SCM is sustainability and ethical practices in the SMEs in the South Asia region. Eco-friendly practices in the supply chain process, such as using eco-friendly materials for packaging, using transport systems that run on renewable energy or bio-fuel. Given the high level of pollution in the South Asian countries, the supply chain network must run entirely based on renewable energy or electric power, which can significantly increase the operational cost of the SCM of the SMEs in South Asia.

1.2. Research Problem

There are various supply chain issues that are persistent and intertwined for the SMEs that try to venture into emerging South Asian economies, thus posing obstacles for effective market entry and growth. These supply chain problems emerge from the infrastructural deficiencies, regulatory environment, and institutional inefficiencies, all of which have a significant impact on the operations of the SMEs because of the relatively limited capabilities and capacities available to the SMEs. In many markets, there are poor transport and logistics facilities that pose delays, higher costs, and less reliability with regard to deliveries. Similarly, fragmented supplies and a lack of visibility of the entire supply chain pose challenges to achieving standards in quality and stock management. Consequently, the main problem centres around how SMEs can formulate resilient, efficient, and transparent supply chain processes.

1.3. Relevance of the Research

The potential in exploring new business ventures in South Asia through entry into the markets of countries like India, Bangladesh, Sri Lanka, and Nepal presents a great scope for development for SMEs, although the lack of an effective supply chain is still a major concern. Current literature tends to examine supply chains from a perspective that caters mostly to large multi-national companies, ignoring the limitations posed by SMEs in terms of resource availability, negotiating power, and use of technology. This paper seeks to address these gaps in current research and provide adaptive solutions.

2. Research Question

2.1. Research Question(s)

RQ1: *What are the primary supply chain challenges faced by SMEs entering South Asian emerging markets?*

RQ2: *How do infrastructure and regulatory environments influence SME supply chain operations in these markets?*

RQ3: *To what extent can digital technologies improve supply chain efficiency and visibility for SMEs?*

RQ4: *What strategies can SMEs adopt to mitigate risks and enhance supply chain resilience in South Asia?*

2.2. Aim of this Study

The aim of this research is to identify the key supply chain challenges faced by SMEs entering emerging markets in South Asia and to identify effective strategies for improving supply chain efficiency, resilience, and market entry success.

2.3. Research Objectives

RO1: *To identify the major supply chain challenges encountered by SMEs in South Asian emerging markets.*

RO2: *To analyse the impact of infrastructural, regulatory, and institutional factors on SME supply chain performance.*

RO3: *To evaluate the role of technology adoption and digitalisation in overcoming supply chain inefficiencies.*

RO3: *To propose strategic recommendations for SMEs to enhance supply chain resilience and competitiveness.*

The research questions and objectives have been developed from the research aim, that is, regarding the identification of the supply chain challenges faced by SMEs entering emerging markets in South Asia and the solutions. In the context of SMEs, infrastructural, regulatory, and institutional are the primary sources of disruption. Furthermore, the technology adaptation and digitalisation are another concern for the SMEs.

3. Literature Review

• *Infrastructure, Market Uncertainty, and Supply Chain Complexity*

The study of Mahmood *et al.* (2025) points to the fact that infrastructural shortcomings are one of the key limitations of SME supply chains in developing markets, particularly in South Asia. Ineffective transport, bad energy supply and insufficient logistics infrastructure lead to longer lead time, more expensive and more uncertainty in operation. Such hinderances in the infrastructure limit the capability of firms to sustain an effective inventory control and limit responsiveness to changes in demand. Furthermore, poor digital infrastructure, including reduced access to high-speed internet, restricts further integration between nodes in the supply chain and, hence, visibility and coordination. According to Cortes-Murcia *et al.* (2022), emerging market tends to underline how the absence of such infrastructures has a rather disproportionate impact on SMEs rather than large corporations, as these organisations have a poor bargaining capacity and cannot afford their own solutions in the area of logistics. In recent years, market uncertainty has gotten worse due to global disruptions, such as

pandemics, geopolitical stress, and variations in demand. The market has been highly fragmented and informal, and in such cases, the demand is more difficult to forecast, as is the case in SMEs in South Asia. As per the study of Tao *et al.* (2025), customer behaviour uncertainties, demand variability, and supplier reliability are some of the most promising supply chain issues that SMEs are confronted with. These uncertainties make planning production and inventory management problematic, which in most cases and situations result in either stockout or over inventory, both of which are detrimental to the financial performance.

SMEs are progressively being involved in global or multi-level supply chains that present challenges in coordination, information asymmetry and dependence threats. The increased interdependence between the suppliers, distributors, and logistics providers poses more vulnerability to disruptions. According to Munna (2025), interruptions in the upstream supply chain, such as slow imports of raw material, can have profound impacts on the operations of SMEs, more so where they do not have alternative sources for their raw materials. Also, the nature of dealing with different stakeholders who have different technological requirements and standards complicates the process of ensuring smooth information flows. Moreover, there is technological turbulence and changing demands of an industry that further complicates things. SMEs are forced to keep changing with the market trends and customer demands, and handle their limited resources. As opined by Chatterjee *et al.* (2023), highly digitalised industries have higher supply chain disruptions, whereby a firm has to update processes and technologies more often. Therefore, limitations in infrastructure, uncertainty in the market, and complexity in the supply chains interplay to produce a difficult operating environment for SMEs and curtail their efficiency and resilience.

- ***Financial Constraints and Resource Limitations***

Financial institutions usually impose restrictions on formal financing of SMEs because of the absence of collateral, poor credit histories and perceived high risk of SMEs. This restricts their ability to invest in supply chain infrastructure, technology, and human capital. The study of Gelsomino *et al.* (2023) indicates that the availability of financing is closely related to the aspects of logistics capability, supply chain relationships, and information transparency, and claims that financially constrained SMEs tend to find themselves in vicious cycles of poor performance. Resource constraints are not limited to financial funds but also to the aspect of human resources and organisational capacity. SMEs usually do not have qualified personnel to oversee a complicated supply chain or use advanced technologies. The other critical concern is the small size of operation, where the SMEs are not able to attain economies of scale.

Smaller-scale units of production add per-unit costs, as well as lower bargaining power against suppliers and logistics suppliers. This causes the SMEs to have higher input costs and unfavourable contractual conditions than in large firms. This drawback is also increased in times of economic crisis, as the strains and stresses of the economy are also increased by the growing cost and declining demand.

The budgets of SMEs are often focused on digital transformation, with most being unsure of whether their investment will translate into revenue and also because of opposing interests within their operations. This leads to decreased rates of adoption and competitiveness in ever-digitised supply chains. External exposure is also increased due to financial vulnerability. SMEs with poor cash stocks are less resilient to shocks like the disruption of supplies, demand shocks, or price shocks in prices. This will cause operational imbalances and, in extreme cases, failure of the business.

- ***Technological and Integration Challenges***

Digitisation of the supply chain has been broadly accepted as one of the main contributors to supply chain efficiency and resilience, but there are severe technological and integration issues with SMEs. Although digitalisation is increasingly becoming important, SMEs are disadvantaged in using modern technologies compared to their bigger counterparts because of financial, technical and organisational challenges. According to Mohna (2025), resource constraints, skills, and non-strategic planning are key drivers behind the failure among SMEs to undergo digital transformation. The inability to have interoperability among various digital systems is one of the most crucial problems. A drawback of SMEs is that they usually have fragmented, old-fashioned legacy systems not compatible with new digital tools, which makes it hard and expensive to integrate. As per Holloway (2025), a lack of integration between systems ensures a lack of visibility and coordination in the supply chain, thus results into inefficiencies and delays. Lack of standardisation within the digital platforms also makes it more challenging to share data and communicate among the key players in the supply chain. Another issue is cybersecurity threats, and the integration of digital technology, like cloud computing and IoT, is making SMEs more exposed to cyber attacks. According to the study of Benjamin *et al.* (2024), data security issues are a source of fear among many SMEs when it comes to the complete adoption of digital change due to the risk of financial loss. Moreover, the expenses of the introduction of secure systems, as well as educating employees on cybersecurity measures, may be resource-prohibitive due to their high cost.

The organisational resistance and cultural barriers also hinder the technological adoption. Lack of familiarity with acceptance of new systems or fear of job displacement may make

employees unwilling to accept the new systems. According to Lakshmi *et al.* (2024), during digital transformation initiatives, there is further complexity in training requirements and the necessity to continuously upskill. There are indications of workforce issues, such as the digital skill gap and change resistance; these will slow the implementation of technology among SMEs. Besides, it is not always a straightforward case of realising the benefits of advanced technologies, which results in uncertainty of return on investment. Such uncertainty deters SMEs from making significant investments in digital infrastructure. Even though new technologies are associated with AI, blockchain, and digital twins can dramatically increase the efficiency and transparency of supply chains, their adoption is costly in terms of financial and technical efforts. Technological and integration issues are therefore a significant issue that has hindered the enhancement of SMEs' supply chain in emerging markets.

- ***Regulatory, Institutional, and Sustainability Pressures***

In determining SME supply chain performance in developing markets, regulatory and institutional aspects are important. Complex and, in many cases, inconsistent systems of regulation pose a high compliance cost to SMEs. Such policies are customs policies, tax policies, and trade policies, which are likely to raise operational costs and make buying and selling across borders painfully slow to transact. These factors are further complicated in South Asia by bureaucratic inefficiency and lack of transparency, which means that these SMEs have difficulties navigating the regulatory environment. The institutional weaknesses that delay the development of the SMEs include poor government support and poor implementation of the policies. Although different policies are put in place to facilitate the growth of SMEs, their effectiveness is limited by the absence of access to information and the lack of a good coordination strategy. SMEs are often not aware of the support programs available or have problems accessing them because of administrative obstacles. This gap in institutions decreases their capacity to lift the supply chain capacities and compete in the international markets.

Another key driver of supply chains has been the sustainability pressures, which have taken on a more crucial role in supply chains. The consumers, the governments and international partners are also expecting more from the firms in terms of environmental and social responsibility. In the case of SMEs, these requirements regarding sustainability can be especially hard to fulfil because of the lack of resources and potential. Environmental regulations, the use of green technologies and the deployment of sustainable practices are usually accompanied by large investments, which many SMEs cannot afford. Meanwhile, global supply chains are emphasising transparency and traceability. SMEs will be required to

provide a lot of data concerning sourcing, production processes and the effect it has on the environment. Nevertheless, to ensure this transparency, there must be high-level information systems and data management systems available, and that is not available to most SMEs. A lack of effective data-sharing platforms and traceability systems constrains the ability of the SMEs to comply with regulatory and sustainability requirements. Moreover, the changing international standards and trade policies also pose more uncertainty to SMEs involved in exports. The constant alteration of regulations may disrupt the supply chain activity and force its constant adjustment. Consequently, the regulatory, institutional, and sustainability pressures cumulatively create huge limitations on SME supply chains, not only in operational efficiency but also in long-term competitiveness.

- ***Conceptual Framework***

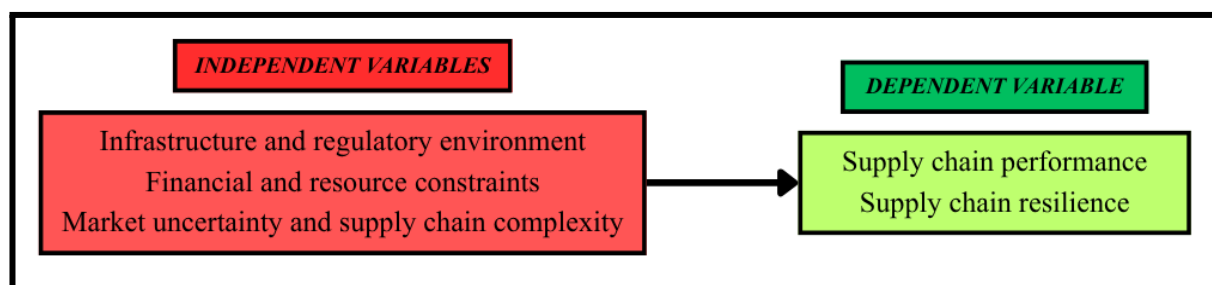


Figure 1: Conceptual Framework

(Source: Self-created)

These independent variables, along with others, are a reflection of the external and internal pressures that form supply chains in the SMEs in South Asia. The operating environment is defined by infrastructure and regulatory conditions, which impact the level of efficiency and compliance costs. The financial and resource constraints restrict the capabilities of SMEs to be strategic in their response, and market uncertainty and complexity of the supply chain expose them to disruption. These aspects have a direct impact on the dependent variables, composed of the supply chain performance and resilience, as they inhibit efficiency, flexibility, and recovery capability. Good infrastructure and favourable legal provisions can contribute to improved results and resource constraints and lack of confidence decrease the influence. In general, the framework indicates that structural issues and the capability of the firms to produce, in turn, define the effectiveness and adaptability of the SME supply chain.

4. Methodology

4.1. Approach

The research approach is a comprehensive plan for a specific study that outlines the way the research will progress, which influences data collection method, data analysis processes, interpretation of the output of the analysis and others (Mulisa, 2022). Inductive and deductive approaches are the two most significant approaches in research, and both are applied in different ways and conception.

An inductive research approach is a bottom-up approach that identifies specific meaning from a dataset collected from observation of specific factors. It generates patterns and themes from collected data and generates new theories that are aligned with the research objectives and can address the research questions (Kumar and Ujire, 2024). This approach is specifically applicable if limited existing data is available and new insights are expected from the research towards a generalised idea. On the other hand, the deductive research approach is a top-down method that either proves or falsifies two or more hypotheses identified before the data collection and analysis process begins. In this approach, the data are specified according to the research aim to test the hypotheses developed, which are usually called the positive hypothesis and the null hypothesis (Yuwono and Rachmawati, 2023). An inductive research approach is heavily related to the descriptive data and context, while the deductive approach focuses on the numerical figures mostly.

In this specific research, the inductive research approach will be implemented by the researcher. The aim of the research is not to achieve specific numerical aspects, but to identify the strategies suitable for the SMEs in South Asia. Furthermore, the researcher will not imply any hypotheses for proving or falsifying. Thus, the choice of this research approach is justified in this study.

4.2. Philosophy

Research philosophy is a set of beliefs and assumptions about the knowledge that is being interpreted and acquired by the researcher in the context of the specifically constructed research objectives. The key aspects of research philosophy are ontology, epistemology, and axiology. Ontology identifies whether a specific reality is objective and external or subjective and socially constructed. On the other hand, epistemology ensures that the sources provide sufficient knowledge from valid and authentic sources (Sun and Zuo, 2024).

Interpretivism and positivism are the two primary research philosophies that are different in nature. Interpretivism is a socially constructed aspect with a subjective viewpoint. The goal of

interpretivism philosophy is to identify the context of a specific phenomenon that answers the ‘why’ and ‘how’ aspects relevant to the research aim (Junjie and Yingxin, 2022). On the other hand, the positivism philosophy has an objective point of view, and it only considers structured and measurable facts. A positivism philosophy helps the researcher to collect data from a sample and apply the implications to a broader aspect (Bergmann, 2023). In terms of the key aspects, for interpretivism philosophy, the ontological, epistemological, and axiological strategies are relativism, subjectivism and value-laden, respectively. On the other hand, the same for the positivism philosophy are realism, objectivism and value-free, respectively.

For this specific research, the researcher will adopt the interpretivism philosophy, as the understanding of the output of the research is subjective and is not measurable through statistical measurement processes. The researcher will interpret the analysed data to address the research questions and meet the research aim; thus, the choice of the research philosophy is justified for this study.

4.3. Strategy & Methods

Research strategy is a systematic plan that provides the researcher with a guideline regarding the way the research will progress. It acts as a bridge between the theoretical approaches and the practical implementations of various aspects of the study to ensure that the research remains productive, efficient and aligned to the aim. As the research design, the exploratory research design will be implemented in this case. An exploratory research design is a flexible design that is used for addressing the research questions whose answers are undefined and or lack in-depth clarity (Olawale *et al.*, 2023). It is primarily tied to qualitative data, focusing on the depth and context of an event. In contrast to this, the explanatory research design is highly structured and focuses on the cause-and-effect of a specific phenomenon to establish the relationship between the variables (Thomas and Zubkov, 2023). It is mostly attached to quantitative data and objectification of data, and thus, is not applicable to this study. There are several research strategies that can be applied in a research process, such as experiment, case study, ethnography, action research and others. For this specific research, the case studies will be used.

In the case of this research, qualitative data will be collected from secondary sources that are authentic and/or peer-reviewed. Qualitative data is descriptive in nature, highly symmetric and flexible in nature and includes the context of a specific research topic instead of the value of the variables only, unlike quantitative data (McBeath, 2022). In the case of quantitative data, the researcher collects the value of the variables only and manipulates the independent

variable to identify the impact on the dependent variables to test the hypothesis developed before collecting and analysing the data, without considering the context of the processes.

The time horizon of the data will be cross-sectional, as the data will be collected within the time span of the last 5 years. To completely understand the challenges of SMEs' SCM in South Asia requires a study that captures the details of more than 5 years; however, the cross-sectional study will reflect the current position of the supply chain network in South Asia, which will provide an idea regarding the issues which are significant at present.

4.4. Sample

Primary data and secondary data are the two types of data that are used in research. Primary data are collected directly from a selected number of people, called samples, while the secondary data are collected from pre-published sources. Semi-structured interviews and structured surveys are the most popular methods of collecting qualitative and quantitative primary data, respectively (Ajayi, 2023). Primary data are highly relevant to the topic of the research, up-to-date, and allow a higher degree of ownership over the collected data. However, primary data collection requires a high amount of resources and time and has a limited scope to generalise and implement to a large population. Apart from these, collecting primary data from preferred respondents often becomes impossible due to the limited reach of the researcher and less knowledge of the common people about a very specific subject. On the other hand, secondary data takes very little time and cost compared to primary data, and the data are highly accessible to the researcher. Also, the researcher can explicitly choose the data sources based on their relevance to the topic. However, the researcher may not know exactly how the data was collected by the author of the secondary sources, which may create ethical dilemmas, and also the researcher often does not have complete control over the secondary data, which may create issues during the usage in the present study.

In this research, the data will be collected from secondary sources that include, but are not limited to, peer-reviewed journal articles, latest newspaper reports, academic books and reputed websites. The journal articles will be collected from sources such as *Researchgate*, *Springer*, *Sciencedirect*, and *MDPI* for the research. On the other hand, for newspaper reports, sources such as *Reuters*, *BBC*, *NYTimes* and *Harvard Business Review* will be used. Moreover, reputed websites such as *Statista* and *Yahoo Finance* will be considered for collecting the data. Also, government websites will contribute to the data set for this research. The sources will not be published before 2022, and they must be in the English language. Also, the core topic of the sources must be aligned with the aim of this research to maintain relevance.

The standard ethical aspects will be implemented during and after the research process. Before the data collection process, permission from the university authorities will be acquired. Apart from that, the collected data will be stored in the university's cloud storage for further use for academic purposes only, as it will not be accessible to any third party. The *UK General Data Protection Regulation* and the *Data Protection Act 2018* will be complied with regarding the data (UK Government, 2025). All the authors of the secondary data will be referenced throughout the research process. However, as there is no human respondent involved in this study, there is no requirement to collect any consent form.

4.5. Analysis

In order to analyse the collected data, the thematic data analysis process will be implemented in this research. The thematic data analysis method is a flexible process that is often used in analysing secondary qualitative descriptive data collected from case studies. Thematic data analysis identifies themes and patterns among the collected secondary data, which are used to address the research questions to meet the research aim.

5. Conclusion

5.1. Implications

The proposed study offers significant benefits for SMEs and the wider business sector by providing a clearer understanding of the supply chain challenges faced when entering emerging markets in South Asia. By identifying key barriers such as infrastructural inefficiencies, financial limitations, and fragmented supply networks, the study enables SMEs to make more informed strategic decisions. This can lead to improved supplier selection, better inventory management, and more efficient logistics planning. Additionally, the research can support SMEs in developing resilience strategies to manage uncertainty and disruptions. For larger firms and multinational corporations, the study highlights the constraints faced by SME partners, encouraging more inclusive supply chain practices and stronger collaboration. As a result, the overall efficiency and integration of supply chains within the region can improve, fostering competitiveness and sustainable business growth across industries.

From a policy and societal perspective, the study has important implications for economic development and social well-being in South Asia. SMEs contribute significantly to employment generation and income distribution, particularly in developing and semi-urban regions. By addressing supply chain inefficiencies, the research can indirectly support job creation, enhance productivity, and improve market access for small producers. Policymakers

can use the findings to design targeted interventions, such as improving infrastructure, simplifying regulatory frameworks, and expanding access to finance for SMEs. Furthermore, the study emphasises the importance of sustainable supply chain practices, encouraging environmentally responsible business operations. This can contribute to reduced environmental impact and more ethical sourcing practices. Overall, the research not only advances academic understanding but also promotes inclusive economic growth, strengthens industrial development, and supports long-term societal benefits in emerging South Asian markets.

5.2. Recommendations

The recommendations for achieving more accurate results are as follows:

- Adopting mixed-method and longitudinal research designs to capture evolving SME supply chain dynamics
- Conducting sector-specific and cross-country comparative studies within South Asia
- Examining the role of informal networks and relationships in supply chain resilience
- Investigating the impact of emerging technologies such as AI, blockchain, and digital platforms on transparency and coordination
- Developing context-specific frameworks integrating infrastructure, institutional, and technological factors
- Exploring barriers to digital adoption and strategies to overcome resource constraints in SMEs
- Implementing gradual digital transformation using cost-effective tools such as cloud systems and data analytics
- Strengthening supplier relationships and developing diversified sourcing strategies
- Investing in employee training to enhance digital and supply chain management capabilities
- Collaborating with government agencies, industry bodies, and logistics partners for support and compliance
- Improving risk management through contingency planning and flexible supply chain design
- Adopting sustainable practices to meet regulatory requirements and enhance competitiveness

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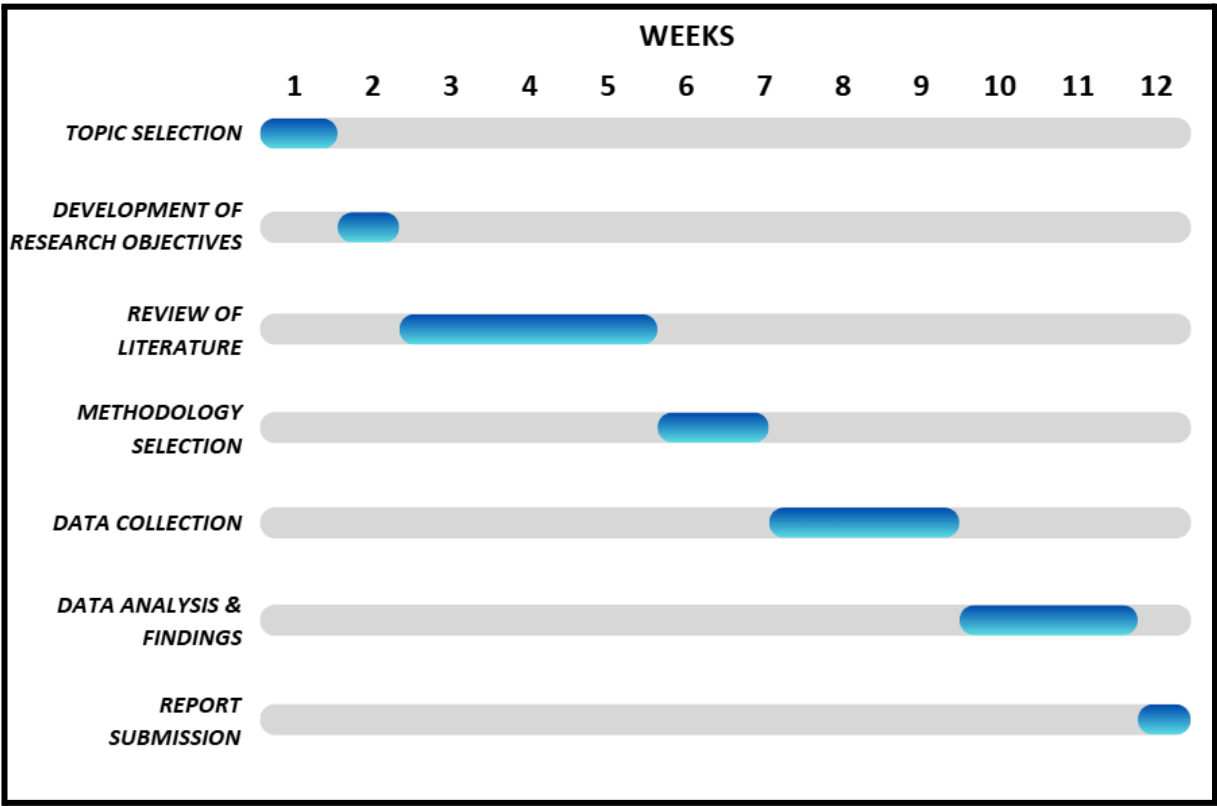
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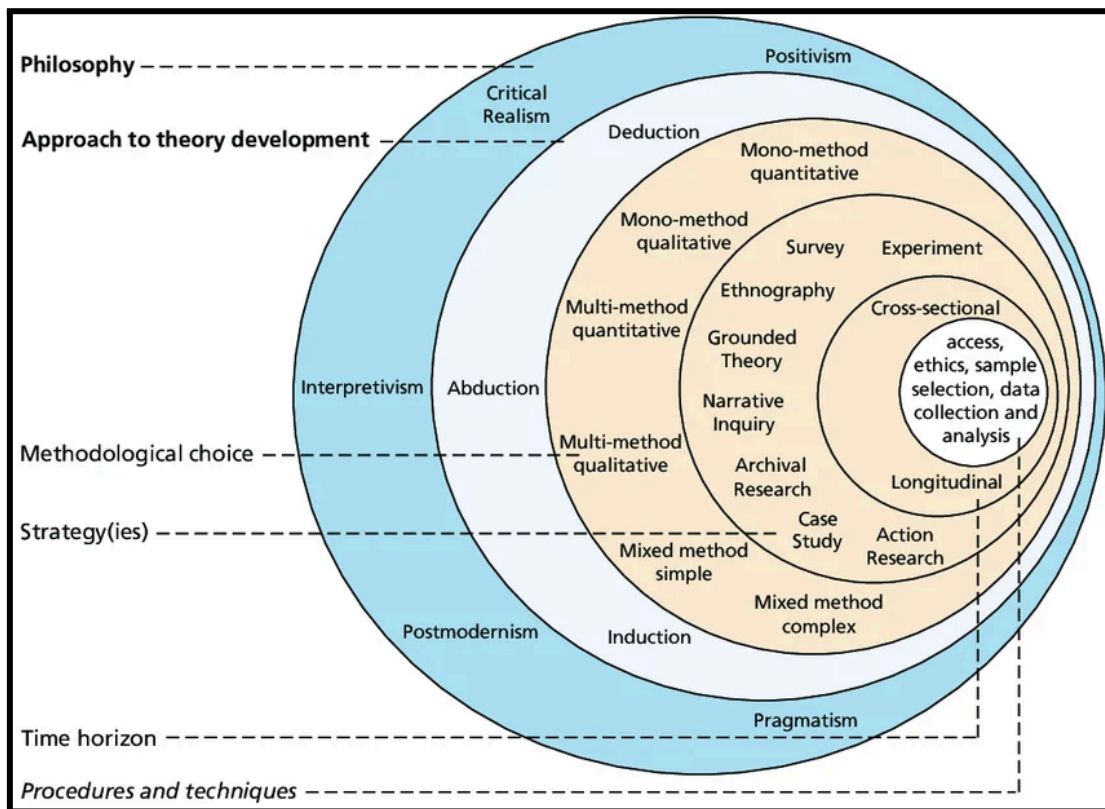
Appendix

Appendix 1: Gantt Chart



(Source: Self-created)

Appendix 2: Research Onion



(Source: Saunders *et al.*, 2023)

